



Infrastructure Asset Management

What is asset management?



Asset Management

An integrated set of processes to minimize the lifecycle costs of infrastructure assets, at an acceptable level of risk, while continuously delivering established levels of service

Infrastructure asset management is the integrated, multidisciplinary set of strategies in sustaining public infrastructure assets such as water treatment facilities, sewer lines, roads, utility grids, bridges, and railways. It focuses the later stages of a facility's life cycle specifically maintenance, rehabilitation, and replacement

Asset Management- Operational Excellence



How do you get more
out of the same resources?



Grow your productivity without growing costs. Talk to us and find out how our capabilities can help you make the most of yours.

• Consulting • Technology • Outsourcing


accenture
High performance. Delivered.

Managing infrastructure assets is nothing new



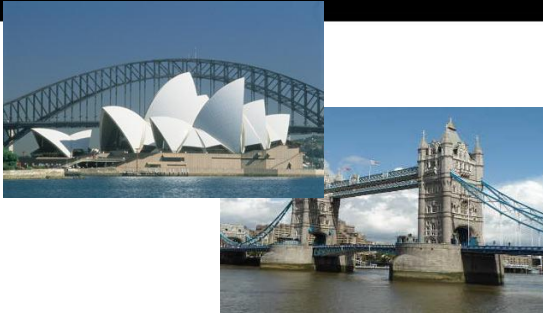
*8th and 9th Century -
Rome's abandoned
aqueducts were
rehabilitated and reused*



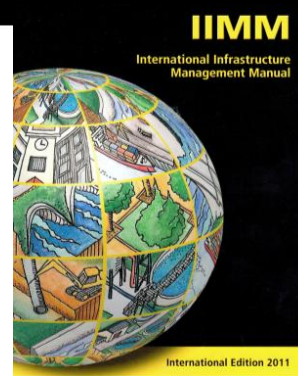
*14th Century – Constitution of
Siena, Italy required maintenance
the City's fountains and other
public works*



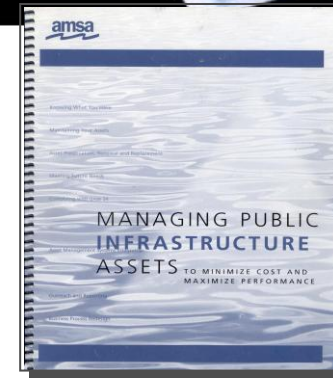
Evolution of modern Asset Management



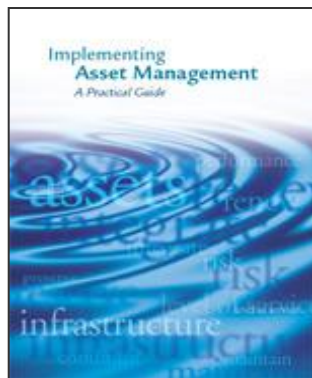
Restructuring of water sector
in Australia and the UK



Comprehensive
International Guidance



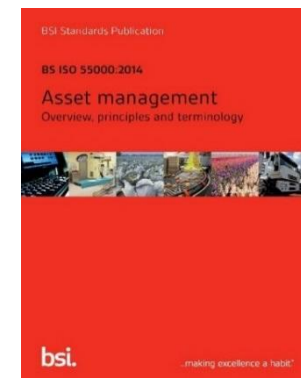
Comprehensive U.S.
Guidance



*The Risk-Based
Approach*



UK Asset Management
Specification (PAS 55)



*New International Standard for
Asset Management
(ISO 55000)*

Asset Management has also become more strategic over the last several years



1st Generation AM *Asset Information*

What do we have?
Where is it?
What condition is it in?
How does it perform?
Maintenance Management

2nd Generation AM *Asset Strategies*

Optimization
Risk-based
System controls
Failure mode analysis
Reliability-centered maintenance

3rd Generation AM *Total Integration*

Coordinated strategy
Service standards
IT integration
Financial analysis
Organizational alignment

Today, Asset Management focuses on integrated processes and tools to facilitate better decisions

- Minimizing lifecycle costs
- Establishing service levels
- Evaluating, understanding and managing risk
- Fact-based decision-making for maintenance, rehabilitation and replacement
- Optimization of operations and maintenance
- Involving collaboration among staff and multi-disciplinary activities
- Consistency across the enterprise

Asset Management

An integrated set of processes to minimize the lifecycle costs of infrastructure assets, at an acceptable level of risk, while continuously delivering established levels of service

Implementing Asset Management provides benefits to both internal and external stakeholders



Benefits

- Informed and defensible asset investment decisions
- Better managed risk
- Improved asset reliability
- Improved efficiency and effectiveness
- Improved service delivery
- Improved customer satisfaction
- Improved regulatory compliance
- Improved bond rating
- Improved internal coordination and communication

Why do we need asset management?



- Existing infrastructure is ageing
- Increased demand for better roads, bridges, sidewalks, lights and improved sewer & water systems
- Higher standards for safety & health
- Environmental protection concerns
- Regulations
- Growth

What are the benefits of asset management?



- ▶ Facilitates the establishment of policy objectives & related measurement of performance
- ▶ Avoids problems & potential crisis
- ▶ Provides better & consistent levels of service for public
- ▶ Reduces life cycle costs

Cont..



- ▶ Allows for better decisions regarding resource allocation
- ▶ Reduces risk of Government
- ▶ Allows for more effective financial planning
- ▶ Leads to more efficient data management
- ▶ Results in positive institutional change

What are the key principals?



1. Strategic & proactive in approach.
2. Places a premium on data, information, collaboration and interdisciplinary management.
3. A long term and comprehensive view of infrastructure performance and cost.
4. An explicit, visible, and transparent approach requiring effective communication among all stakeholders.
5. A business case involving investment choices that are policy driven with tradeoffs among competing priorities.

What are the essential components of the plan?



1. Asset Value
2. Life Cycle Management
3. Sustainability
4. Integration of Technical & Financial Plans
5. Risk Assessment
6. Performance Management

1. Asset Value



- It must be recognized that all assets have a monetary value. This value is used to determine the depth of re-investment for each asset. By knowing the “value” the type of rehab strategy and/or optimum replacement time can be determined.
- Finance Dept is determining actual value of assets through a capital cost study.

2. Life Cycle Management

- All assets have a finite life expectancy.
- ✓ Estimate rate of deterioration
- ✓ Decision point – what impact will work done at any point in the life cycle have on remaining asset life, operation costs and other related expenses.



3. Sustainability



- “Meeting the needs of the present generation without compromising the ability of future generations to meet their own needs.” (National Guide to Sustainable Municipal Infrastructure)
- Current users pay a fair share for the service they receive so that future users do not have to pay a higher cost for the same level of service.

Technical & Financial Plan Integration

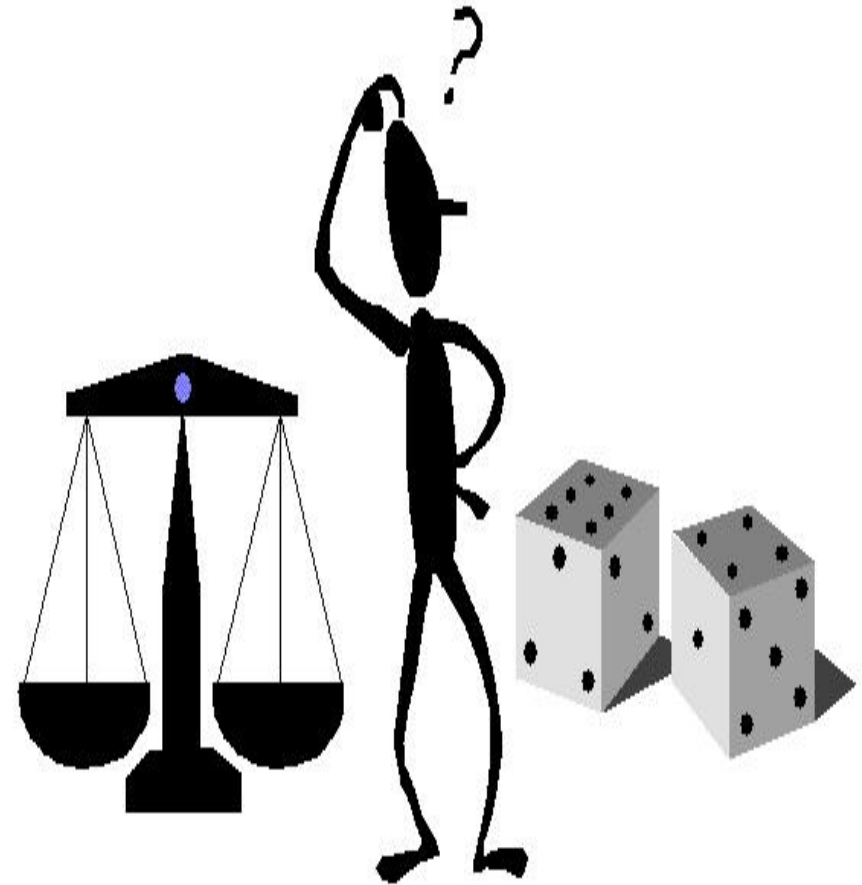


Financial Plan	Technical Plan
•Identify financial investment required per year for each asset's life cycle over the long term •Identify exceptional expenditures required to meet requirements of technical plan based on condition surveys	•Minimize life cycle costs •Maintain adequate level of service •Maintain service level at lowest possible risk
Cash Flows	

5. Risk Assessment



- Acceptable risk tolerance for each asset must be part of the strategy
- Condition survey used to determine rate of failure and consequences of failure
- Risk Factors include: financial, environmental, regulatory/legal, public health and safety



6. Performance Measurement

We need to measure what we do



- ❖ Monitor strategies regularly
- ❖ Make adjustments at the right stage of the asset's life cycle to achieve the balance between cost and level of service
- ❖ use benchmarks to determine performance of assets

Case Study : Panda Brandywine power plant



- In 1993, Panda Energy, the predecessor organisation to the Panda Power Funds, began work on a combined cycle gas turbine (CCGT) power plant.
- Last 25 years, it created value in more than 9,000 megawatts of power generation facilities built at a cost of approximately \$6 billion.
- IRR of more than 70 percent and a multiple of 17 times money. Altogether, Panda Energy's \$13.9 million investment has realised, to date, more than \$236 million

Key drivers for value creation in a power plant



- **Market-driven business development**
- **Market selection** (with a growing demand for electric power, long term PPAs)
- **Strategic site** (close proximity to the physical resources necessary to operate the power station.
Specifically, natural gas pipelines, transmission lines, a nearby substation, an adequate supply (and type) of water and sufficient water disposal areas)
- **Staged, cost-controlled project development** (fatal flow early on in the development process, so that limited investment is put at risk)
- **Proactive community relations**
- **A well-rounded, experienced team**
- **Financially sound counterparties** (Pepco, the project's power purchaser; General Electric, which provided equipment and financing for the project; Raytheon, which constructed Brandywine; and Detroit Edison (DTE) which supplies natural gas to the project)
- **Tight contracts**



Figure 7.3: Most value creation occurs during development and construction

Development	Construction		Operation		
Year 1 & 2	Year 3	Year 4	Year 5	Year 6	Year 7
Risks	Risks		Risks		
• Community support / tax agreements • Electric transmission • Power purchase agreements • Environmental permits • Site / land • Gas Supply / transport • Water Supply / disposal • Project Debt / equity financing	• Change in prices • On time completion • Equipment / contractor performance • Equipment reservation • Force majeure and property / casualty		• Electric pricing • Fuel pricing • Regulatory • Operating performance / cost structure • Property / casualty / business interruption		
Mitigants	Mitigants		Mitigants		
• Fatal flaw analyses • Strict cost control • Experienced development and finance team • Limited investment at risk	• Fixed price contracts • Contractor schedule and performance guarantees with liquidated damages • Contingencies • Warranties from vendors • Insurance • Experienced construction team • Aggressive inspection / oversight		• Long-term power contracts with fixed capacity pricing; pass through of all variable costs including fuel, regulatory and O&M • Experienced operating and management team • Excellent track record of high availability • Insurance		



- **Operational Excellence-** capacity-based revenue model and an asset management strategy achieved by utilising reliable equipment, plant designs that include proper redundancy and durability, and best-in-class management information systems
- **Finance –**
 - Panda Energy financed the Brandywine project with a \$215 million non-recourse construction loan from GE Capital for 20-year term and two five-year renewal options
 - In July 1996, Panda issued \$105.5 million pooled project bonds through a subsidiary funding corporation
 - The pooled bonds were also supported by cash flows from Panda's 180-megawatt Rosemary facility
 - In 1997, Panda raised another \$155 million worth of bonds based on cash flows from Brandywine and other projects to fund additional development efforts. The bonds were subsequently repaid in 2001.
 - Northland Power Income Fund acquired a 19 percent ownership interest in the Brandywine and Rosemary facilities in return for a \$93 million loan and a cash payment of \$6 million



Thank you !